

MUHAMMED MUIZ ARUMMAL

AI Engineer

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PROFILE

AI engineer who builds and evaluates LLM and multi-agent systems — from the data pipeline through to a deployed product. MSc AI & Machine Learning at the University of Birmingham, based in the UK. I focus on whether an AI system actually works in the real world, not just in a demo: building the product and the tests that prove it.

EDUCATION

University of Birmingham — MSc, Artificial Intelligence & Machine Learning in progress, expected 2027
TKM Institute of Technology — B.Tech, Computer Science & Engineering 2025

RESEARCH

Testing AI attacker agents against a multi-agent app github.com/seclayer/finbot-bench

- Built a benchmark where local LLMs act as autonomous attackers against a deliberately vulnerable multi-agent AI application (**OWASP FinBot CTF**), scored by the app's own exploit detector rather than the attacker marking its own work.
- Compared three attacker designs — one-shot, a feedback loop, and a team of cooperating agents — and found the attacker's *design* mattered more than the size of the model behind it.

Are AI safety scores measuring the right thing? preprint in preparation | github.com/seclayer/jlens-fuzz

- Showed that a widely reused automated judge, which decides whether an attack on a chatbot “worked”, rewards replies that merely *look* like a jailbreak over ones that are actually harmful — so published success rates built on it are inflated.
- Built a stricter replacement judge and reported honestly where it still fails, rather than claiming a clean fix.

Independent reproduction of a 2026 jailbreak paper github.com/seclayer/reproductions

- Rebuilt the paper's method end to end on three local models, and tested a defence the original authors named but never evaluated.
- Caught the automated grader scoring wrongly in both directions, re-checked every result by hand, and published the corrections next to the originals.

Security audit of a deployed multi-agent platform authorization review & fix

- Reviewed a live agent-to-agent platform against a published threat checklist, found a missing access-control check, fixed it, and proved the fix with an automated test.

PROJECTS

Multi-agent AI assistant platform (GptGram) github.com/seclayer/Gptgram-production-ready-

- Built and deployed an AI platform on **AWS** where several **AI agents** work together to answer a question, with a **retrieval (RAG)** layer on **Qdrant** so answers are grounded in real documents.
- Ran it under real load with background job queues (**Celery**), so many people can use it at once without slowing down.

AI system for security monitoring (Secure Guard) github.com/seclayer/ai-soc

- Built a data pipeline (**Python, SQL**) that turns messy raw security logs into clean data, and trained **machine-learning** models to flag likely malware so analysts see real threats first.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL.
- **AI / ML:** LLMs, multi-agent & agentic systems, RAG, model evaluation, PyTorch, TensorFlow, JAX, scikit-learn.
- **Cloud & data:** AWS, Docker, Git / CI-CD, REST APIs, Databricks, Power BI.
- **Tools / frameworks:** LangChain, Qdrant, Ollama.
- **Security background:** detection engineering, SIEM (ELK, Splunk), MITRE ATT&CK.

CERTIFICATIONS

- **CompTIA CySA+** (Cybersecurity Analyst) — [verify](#) | **Mathematics for Machine Learning**, Imperial College London (Coursera) — [verify](#)